

Math 271: Mathematical Methods

The Inverse Function Theorem in one variable

The statement

Inverse Function Theorem

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be C^1 and suppose $f'(x_0) \neq 0$ for some $x_0 \in \mathbb{R}$. Then there exist open intervals $U \ni x_0$ and $V \ni f(x_0)$ such that

- $f : U \rightarrow V$ is a bijection (so $f^{-1} : V \rightarrow U$ exists);
- $f^{-1} : V \rightarrow U$ is C^1 ;
- for every $y \in V$, $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$.

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We assume $f'(x_0) > 0$. (For $f'(x_0) < 0$, replace “increasing” by “decreasing”.)

(a) A positive lower bound for f' near x_0

Claim

There exist $\delta > 0$ and $c > 0$ such that $f'(x) \geq c > 0$ for all $x \in (x_0 - \delta, x_0 + \delta) \equiv U$.

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Proof.

f' is continuous at x_0 . Put $c := \frac{1}{2}f'(x_0) > 0$ and use $\varepsilon = c$: there is $\delta > 0$ such that

$$x \in U \implies |f'(x) - f'(x_0)| < c. \quad (1)$$

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Hence for every $x \in U$: $f'(x) > f'(x_0) - c = 2c - c = c$. □

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Hence for every $x \in U$: $f'(x) > f'(x_0) - c = 2c - c = c$. □

In particular $f'(x) \neq 0$ for every $x \in U$.

(b) f is strictly increasing on U

Claim

For $x_1, x_2 \in U$ with $x_2 > x_1$, we have $f(x_2) > f(x_1)$.

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Proof.

Let $x_1 < x_2$ in U ; then $[x_1, x_2] \subset U$. By the MVT there is $\xi \in (x_1, x_2) \subset U$ with

$$f(x_2) - f(x_1) = f'(\xi)(x_2 - x_1).$$

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$$f(x_2) - f(x_1) = f'(\xi)(x_2 - x_1).$$

By (a), $f'(\xi) \geq c$, and $x_2 - x_1 > 0$, so

$$f(x_2) - f(x_1) \geq c(x_2 - x_1) > 0. \quad (*)$$



(c) A lower Lipschitz bound for f

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Case $x_1 < x_2$: by (*) in (b), $f(x_2) - f(x_1) \geq c(x_2 - x_1) > 0$. Both sides are positive, so

$$|f(x_2) - f(x_1)| = f(x_2) - f(x_1) \geq c(x_2 - x_1) = c|x_2 - x_1|.$$

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Consequence: $f(x_1) = f(x_2) \Rightarrow c|x_1 - x_2| \leq 0 \Rightarrow x_1 = x_2$, so f is injective on U .

(d) $V = f(U)$ is an interval

Claim

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Step 1: $x_1 \leq x_2$. Otherwise $x_2 < x_1$, and by (b) $y_2 = f(x_2) < f(x_1) = y_1$, contradicting $y_1 \leq y_2$.

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Step 1: $x_1 \leq x_2$. Otherwise $x_2 < x_1$, and by (b) $y_2 = f(x_2) < f(x_1) = y_1$, contradicting $y_1 \leq y_2$.

Step 2: *IVT.* $[x_1, x_2] \subset U$ and f is continuous on $[x_1, x_2]$ with $f(x_1) = y_1 \leq z \leq y_2 = f(x_2)$. So there is $x \in [x_1, x_2] \subset U$ with $f(x) = z$, i.e. $z \in V$. □

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Let $y_0 = f(\bar{x}) \in V$ with $\bar{x} \in U$. Since U is open, choose $r > 0$ with $[\bar{x} - r, \bar{x} + r] \subset U$, and put $d := c r > 0$.

- **Step right.** By (c) (the hint), $|f(\bar{x} + r) - y_0| \geq c r = d$; by (b), $f(\bar{x} + r) > y_0$. Hence $f(\bar{x} + r) \geq y_0 + d$.

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- **Step left.** Likewise $|f(\bar{x} - r) - y_0| \geq d$ and $f(\bar{x} - r) < y_0$, so $f(\bar{x} - r) \leq y_0 - d$.

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- **Step left.** Likewise $|f(\bar{x} - r) - y_0| \geq d$ and $f(\bar{x} - r) < y_0$, so $f(\bar{x} - r) \leq y_0 - d$.
- **Fill in.** $f(\bar{x} - r)$ and $f(\bar{x} + r)$ both lie in V , and V is an interval by (d), so
$$(y_0 - d, y_0 + d) \subset [f(\bar{x} - r), f(\bar{x} + r)] \subset V.$$

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Conclusion

V is open, so by (d) it is an open interval. As f is injective on U (by (b)) and $V = f(U)$, $f : U \rightarrow V$ is a **bijection**; f^{-1} exists.

(f)(i) f^{-1} is Lipschitz on V

Claim

$$|f^{-1}(u) - f^{-1}(v)| \leq K |u - v| \text{ for all } u, v \in V, \text{ with } K = \frac{1}{c} > 0.$$

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Proof.

Let $u, v \in V$ and put $x_1 = f^{-1}(u)$, $x_2 = f^{-1}(v) \in U$, so $f(x_1) = u$, $f(x_2) = v$.

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$$|u - v| = |f(x_1) - f(x_2)| \geq c |x_1 - x_2| = c |f^{-1}(u) - f^{-1}(v)|.$$

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$$|u - v| = |f(x_1) - f(x_2)| \geq c |x_1 - x_2| = c |f^{-1}(u) - f^{-1}(v)|.$$

Dividing by $c > 0$: $|f^{-1}(u) - f^{-1}(v)| \leq \frac{1}{c} |u - v|$. □

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Proof.

Let $v \in V$ and $\varepsilon > 0$. Take $\eta = c\varepsilon > 0$. If $u \in V$ and $|u - v| < \eta$, then by (f)(i)

$$|f^{-1}(u) - f^{-1}(v)| \leq \frac{1}{c} |u - v| < \frac{1}{c} \cdot c\varepsilon = \varepsilon.$$

Hence f^{-1} is continuous at every $v \in V$. □

(g) Differentiability of f^{-1} : the difference quotient

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For every $y \in V$, f^{-1} is differentiable at y and $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$.

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- $x_1 \neq x$, because f^{-1} is injective (it has the inverse f).
- $f(x_1) = y_1$ and $f(x) = y$, so

$$\frac{f^{-1}(y_1) - f^{-1}(y)}{y_1 - y} = \frac{x_1 - x}{f(x_1) - f(x)} = \frac{1}{\frac{f(x_1) - f(x)}{x_1 - x}}.$$

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So the difference quotient of f^{-1} at y is the *reciprocal* of the difference quotient of f at x .

(g) continued: passing to the limit

Let $Q(t) = \frac{f(t) - f(x)}{t - x}$ for $t \in U$, $t \neq x$. Since f is differentiable at x ,
 $Q(t) \rightarrow f'(x)$ as $t \rightarrow x$.

- As $y_1 \rightarrow y$: $x_1 = f^{-1}(y_1) \rightarrow f^{-1}(y) = x$ by continuity of f^{-1} ((f)(ii)), and $x_1 \neq x$.

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- $f'(x) \geq c > 0$ by (a), so $f'(x) \neq 0$ and

$$\frac{f^{-1}(y_1) - f^{-1}(y)}{y_1 - y} = \frac{1}{Q(x_1)} \longrightarrow \frac{1}{f'(x)} \quad (y_1 \rightarrow y).$$

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f^{-1} is differentiable at y and $(f^{-1})'(y) = \frac{1}{f'(x)} = \frac{1}{f'(f^{-1}(y))}$.

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Conclusion

f^{-1} is differentiable at y and $(f^{-1})'(y) = \frac{1}{f'(x)} = \frac{1}{f'(f^{-1}(y))}$.

Quantitatively: by (f)(i), $|x_1 - x| \leq \frac{1}{c}|y_1 - y|$, so $x_1 \rightarrow x$ at least as fast as $y_1 \rightarrow y$.

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Proof.

By (g), $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$ for $y \in V$.

- $f^{-1} : V \rightarrow U$ is continuous by (f)(ii), and $f' : U \rightarrow \mathbb{R}$ is continuous ($f \in C^1$);



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Therefore $(f^{-1})'$ is continuous on V , i.e. f^{-1} is C^1 on V . □

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With $U = (x_0 - \delta, x_0 + \delta)$ from (a) and $V = f(U)$:

- | | |
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| (b) f strictly increasing on U | MVT |
| (c) $ f(x_2) - f(x_1) \geq c x_2 - x_1 $ | from (*) in (b) |
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This proves the theorem when $f'(x_0) > 0$. For $f'(x_0) < 0$, apply it to $-f$; equivalently, apply it to $f/f'(x_0)$, which handles both signs at once.

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- So the hypothesis $f'(x_0) \neq 0$ is exactly what is needed for a differentiable inverse.